

Micro-lab ticket

Computational cost — Friday, October 30

Name: _____ Language (circle): Octave / Python

New syntax (fill as you go)

I needed to...	Command / pattern (your language)
Outer product ab^T	
Inner product $b^T c$	
Time a block of code	
Norm of a difference	

Same math: $(ab^T)c = a(b^T c)$. Quiz 8 will cover Week 10 (products / powers / cost).

E1. $a = (1, 0, -1)$, $b = (2, 1, 0)$, $c = (1, 1, 1)$. Both orders.

Expected: $(3, 0, -3)$. **Yours:** _____

E2. Storage for general n : entries in ab^T ? In scalar $s = b^T c$? For $n = 2000$?

Expected: n^2 vs 1; 4,000,000 vs 1. **Yours:** _____

E3. Random $n = 500$: report `norm(y1-y2)`.

Expected: tiny ($\sim 10^{-10}$ or smaller). **Yours:** _____

E4. Times (seconds) — fill what you can:

n	Order 1 $(ab^T)c$	Order 2 $a(b^T c)$
200		
1000		
3000		

E5. For $n = 10^4$, which order? Two quantitative reasons (storage + timing):

Stuck on (optional): E_____

Instructor initials: _____